

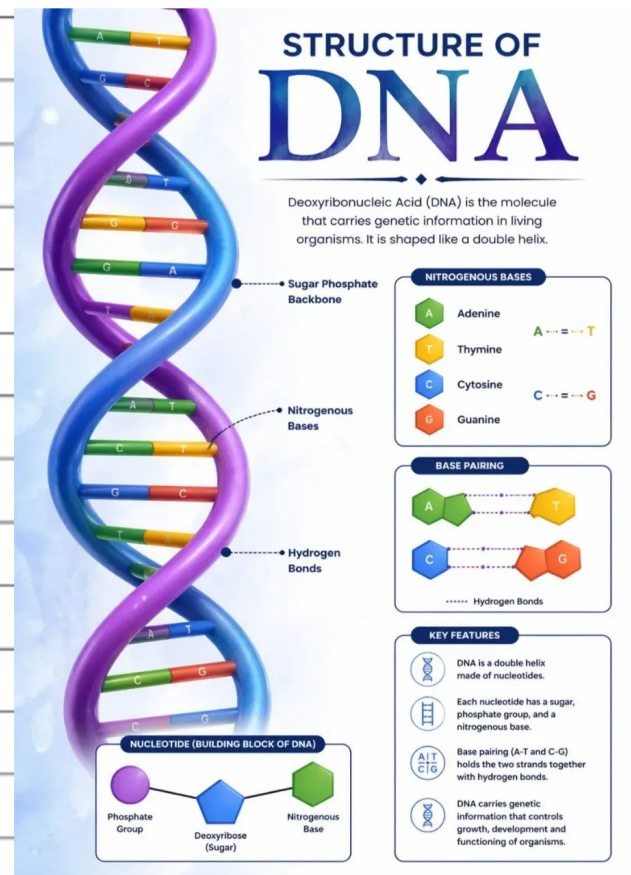
Topic:-

Prepare a model of the DNA and give a presentation based on it.

Introduction:-

In this project we are going to discuss about DNA and its Discovery, its structure, its function, DNA Replication, Chargaff's Rule and Diagram. We will discuss that what is DNA and its types.

♣ What is DNA?



DNA is a group of molecules that is responsible for carrying and transmitting hereditary material or genetic instructions from parents to offspring. This is also true for viruses, small of these entities have either DNA or RNA as their genetic material. For instance, some viruses may have RNA as their genetic material, while others have DNA as their genetic material. The virus HIV contains RNA, which is then converted into DNA after attaching itself to the host cell. Apart from being responsible for the inheritance of genetic information in all living beings, DNA also plays a crucial role in the production of proteins. Nuclear DNA is the DNA contained within the nucleus of every cell in a eukaryotic organism. It codes for the majority of the organism's genome, while the mitochondrial DNA and plasmid DNA handle the rest.

❖ Full form of DNA:-

DNA is known as Deoxyribonucleic Acid. It is an organic compound that has a unique molecular structure. It is found in all prokaryotic cells and eukaryotic cells.

❖ DNA Types:-

There are three different DNA types:

A-DNA: It is a right-handed double helix similar to the B-DNA form.

Dehydrated DNA takes an A form that protects the DNA during extreme conditions such as desiccation. Proteins binding also removes the solvent from DNA and the DNA takes on A form.

B-DNA: This is the most common DNA conformation and is a right-handed helix. The majority of DNA has a B type conformation under normal physiological conditions.

Z-DNA: Z-DNA is a left-handed DNA where the double helix winds to the left in a zig-zag pattern. It was discovered by Andres Wang and Alexander Rich. It is found ahead of the start site of a gene and hence is believed to play some role in gene regulation.

❖ Who discovered DNA:-

DNA was first recognized and identified by the Swiss biologist Johannes Friedrich Miescher in 1869 during his research on white blood cells. The double helix structure of a DNA molecule was later discovered through the experimental data by James Watson and Francis Crick. Finally, it was proved that DNA is responsible for storing genetic information in living organisms.

❖ DNA Replication:-

DNA replication is an important process that occurs during cell division. It is also known as semi-conservative replication during which DNA makes a copy of itself. DNA replication takes place in three stages.

Step 1: Initiation

The replication of DNA begins at a point known as the origin of replication. The two DNA strands are separated by the DNA helicase. This forms the replication fork.

Step 2: Elongation

DNA polymerase III reads nucleotides on the template strand and makes a new strand by adding complementary nucleotides one after the other. For example, if it reads an Adenine on the template strand, it will add a Thymine on the complementary strand. While adding nucleotides to the lagging strand, gaps are formed between the strands. These gaps are known as Okazaki fragments.

Step 3: Termination

The termination sequence present opposite to the origin of replication terminates the replication process. The Tus proteins bind to termination sequence and halt DNA polymerase movement. It induces termination.

❖ DNA function:-

DNA is the genetic material which carries all the hereditary information. Genes are the small segments of DNA consisting mostly of 250 to 2 million base pairs. A gene is a polypeptide molecule where three nitrogenous base sequences stand for one amino acid. Polypeptide chains are further folded in secondary, tertiary and quaternary structures to form different proteins. As every organism contains many genes in its DNA, a large number of proteins can be formed. Proteins are the main functioning and structural molecules in most organisms. Apart from storing genetic information, DNA is involved in further cellular processes.

Replication process:- Transferring the genetic information from one cell to its daughter and from one generation to the next and equal distribution of DNA during the cell division.

- Mutations:- The changes which occur in the DNA sequences.

- Transcription

- Cellular Metabolism

- DNA finger printing
- Gene Therapy

♣ DNA structure:-

DNA is a two-stranded molecule that appears twisted, giving it a unique shape referred to as the double helix.

Each of the two strands is a long sequence of nucleotides. These are the individual units of DNA and they are made of:

- a phosphate molecule
- a sugar molecule called deoxyribose containing five carbons
- a nitrogen-containing region

♣ There are four types of nitrogen-containing regions called bases, including:

- adenine (A)
- cytosine (C)
- guanine (G)
- thymine (T)

The order of these four bases forms the genetic code, which is the instructions for the life.

♣ Chargaff's Rule:-

Erwin Chargaff, a biochemist, discovered that the number of nitrogenous bases in the DNA was present in equal quantities. The amount of A is equal to T, where the amount of C is equal to G.

$$A = T; C = G.$$